

LEVEL VI

SINGLE ANSWER QUESTIONS

1. Let $f : [2, 7] \rightarrow [0, \infty)$ be a continuous and differentiable function. Then, the value of

$$(f(7) - f(2)) \frac{(f(7))^2 + (f(2))^2 + f(2) \cdot f(7)}{3}$$

is (where $c \in (2, 7)$)

- A) $3f^2(c)f'(c)$ B) $5f^2(c) \cdot f(c)$
C) $5f^2(c) \cdot f'(c)$ D) None of these

2. The tangent to the curve $y = e^x$ drawn at the point (c, e^c) intersects the line joining the points $(c-1, e^{c-1})$ and $(c+1, e^{c+1})$

- A) on the left of $x = c$ B) on the right of $x = c$
C) at no point D) at all points

3. If f is continuous function in $[1, 2]$ such that $|f(1) + 3| < |f(1)| + 3$ and $|f(2) + 10| = |f(2)| + 10$, ($f(2) \neq 0$), then the function f in $(1, 2)$ has

- A) at least one root B) no root
C) exactly one root D) none of these

4. The function 'g' defined by $g(x) = f(x^2 - 2x + 8) + f(14 + 2x - x^2)$, where $f(x)$ is twice differentiable function, $f''(x) \geq 0$ for all real numbers x . The function $g(x)$ is increasing in the interval

- A) $[-1, 1] \cup [2, \infty)$ B) $(\infty, -1] \cup [1, 3]$
C) $[-1, 1] \cup [3, \infty)$ D) $(\infty, -2] \cup [1, \infty)$

5. The length of a longest interval in which the function $3 \sin x - 4 \sin^3 x$ is increasing, is [IIT 2002]

- A) $\frac{\pi}{3}$ B) $\frac{\pi}{2}$ C) $\frac{3\pi}{2}$ D) π

6. In $[0, 1]$ Lagrange's Mean Value theorem is not applicable to [IIT 2003]

A) $f(x) = \begin{cases} \frac{1}{2} - x & x < \frac{1}{2} \\ \left(\frac{1}{2} - x\right)^2 & x \geq \frac{1}{2} \end{cases}$

B) $f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1 & x = 0 \end{cases}$

C) $f(x) = x|x|$ D) $f(x) = |x|$

7. Tangent is drawn to ellipse

$$\frac{x^2}{27} + y^2 = 1 \text{ at } (3\sqrt{3}\cos\theta, \sin\theta) \text{ (where } \theta \in (0, \pi/2)).$$

Then the value of θ such that sum of intercepts on axes made by this tangent is minimum, is [IIT 2003]

- A) $\frac{\pi}{3}$ B) $\frac{\pi}{6}$ C) $\frac{\pi}{8}$ D) $\frac{\pi}{4}$

8. If the tangent at (x_1, y_1) to the curve $x^3 + y^3 = a^3$ meets the curve again at (x_2, y_2) then

A) $\frac{x_2}{x_1} + \frac{y_2}{y_1} = -1$ B) $\frac{x_2}{y_1} + \frac{x_1}{y_2} = -1$

C) $\frac{x_1}{x_2} + \frac{y_1}{y_2} = -1$ D) $\frac{x_2}{x_1} + \frac{y_2}{y_1} = 1$

9. If $0 < a < b < \frac{\pi}{2}$ and

$$f(a, b) = \frac{\tan b - \tan a}{b - a}, \text{ Then}$$

- A) $f(a, b) \geq 2$ B) $f(a, b) > 1$
C) $f(a, b) \leq 1$ D) None

10. $f(x)$ is a polynomial of degree 4 with real coefficients such that $f(x) = 0$ is satisfied by $x = 1, 2, 3$ only, then

$$f'(1) \cdot f'(2) \cdot f'(3) \text{ is equal to}$$

- A) 0 B) 2 C) -1 D) None of these

11. If $f(x)$ is a polynomial of degree 5 with real coefficients such that $f(|x|) = 0$ has 8 real roots, then $f(x) = 0$ has

- A) 4 real roots B) 5 real roots
C) 3 real roots D) nothing can be said

12. If the function $f(x) = |x^2 + a|x + b|$ has exactly three points of non-differentiability, then which of the

APPLICATION OF DERIVATIVES

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following can be true ?

- A) $b = 0, a < 0$ B) $b < 0, a \in R$
C) $b > 0, a \in R$ D) All of the above

13. Let f be continuous and differentiable

function such that $f(x)$ and $f'(x)$ have opposite signs everywhere. Then

- A) f is increasing B) f is decreasing
C) $|f|$ is non-monotonic D) $|f|$ is decreasing

14. If $f(x) = 4x^3 - x^2 - 2x + 1$ and

$$g(x) = \begin{cases} \min\{f(t) : 0 \leq t \leq x\}; & 0 \leq x \leq 1 \\ 3 - x & 1 < x \leq 2 \end{cases} \text{ then}$$

$g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right)$ has the value equal to

- A) $7/4$ B) $9/4$ C) $13/4$ D) $5/2$

15. The largest term in the sequence

$$a_n = \frac{n^2}{n^3 + 200} \text{ is given by}$$

- A) $\frac{529}{49}$ B) $\frac{8}{89}$ C) $\frac{49}{543}$ D) $\frac{89}{8}$

16. The number of values of k for which the equation $x^3 - 3x + k = 0$ has two distinct roots lying in the interval $(0, 1)$ is

- A) three B) two C) infinitely many D) zero

17. The maximum value of

$$\left(\sqrt{-3+4x-x^2}+4\right)^2 + (x-5)^2 \text{ (where } 1 \leq x \leq 3) \text{ is}$$

- A) 34 B) 36 C) 32 D) 20

18. On the interval $\left[\frac{5\pi}{4}, \frac{4\pi}{3}\right]$ the least value of

$$\text{the function } f(x) = \int_{5\pi/4}^x (3\sin t + 4\cos t) dt \text{ is}$$

- A) $\frac{3}{2} + \frac{1}{\sqrt{2}} - 2\sqrt{3}$ B) $\frac{3}{2} - \frac{1}{\sqrt{2}} + 2\sqrt{3}$
C) $\frac{3}{2} - \frac{1}{\sqrt{2}} - 2\sqrt{3}$ D) $\frac{3}{2} - 2\sqrt{3}$

19. The function $f(x) = \frac{\ln(\pi + x)}{\ln(e + x)}$

[IIT 1995]

- A) increasing on $(0, \infty)$
B) decreasing on $(0, \infty)$
C) increasing on $\left(0, \frac{\pi}{e}\right)$, decreasing on $\left(\frac{\pi}{e}, \infty\right)$
D) decreasing on $\left(0, \frac{\pi}{e}\right)$, increasing on $\left(\frac{\pi}{e}, \infty\right)$

20. One corner of a long rectangular sheet of paper of width 1 unit is folded over so as to reach the opposite edge of the sheet. The minimum length of the crease is

- A) $\frac{3\sqrt{3}}{4}$ B) $\frac{3\sqrt{3}}{2}$ C) $4\sqrt{3}$ D) $3\sqrt{3}$

21. Let S be a square of unit area. Consider any quadrilateral which has one vertex on each side of S . If a, b, c and d denote the length of the sides of the quadrilateral, then

$$a^2 + b^2 + c^2 + d^2 \text{ lies in}$$

- A) $[3, 5]$ B) $[2, 4]$ C) $[1, 3]$ D) $[0, 2]$

22. The tangent lines for the curve

$$y = \int_0^x 2|t| dt \text{ which are parallel to the}$$

bisector of the first coordinate angle, is given by

- A) $y = x + \frac{3}{4}, y = x - \frac{1}{4}$
B) $y = -x + \frac{1}{4}, y = -x + \frac{3}{4}$
C) $x + y = 2, x - y = 1$
D) $y = x + \frac{1}{4}, y = x - \frac{1}{4}$

23. If $f(x) = x^3 + bx^2 + cx + d$ and $0 < b^2 < c$, then in $(-\infty, \infty)$ [IIT 2004]

- A) $f(x)$ is a strictly increasing function
 B) $f(x)$ has a local maxima
 C) $f(x)$ is strictly decreasing function
 D) $f(x)$ is bounded
24. If $f(x) = x^\alpha \log x$ and $f(0) = 0$, then the value of α for which Rolle's theorem can be applied in $[0, 1]$ is [IIT 2007]
 A) -2 B) 0 C) $1/2$ D) $-1/2$
25. Consider the two curves :
 $C_1 : y^2 = 4x$; $C_2 : x^2 + y^2 - 6x + 1 = 0$,
 Then [IIT 2008]
 A) C_1 and C_2 touch each other only at one point
 B) C_1 and C_2 touch each other exactly at two points
 C) C_1 and C_2 intersect (but do not touch) at exactly two points
 D) C_1 and C_2 neither intersect nor touch each other
26. The total number of local maxima and local minima of the function

$$f(x) = \begin{cases} (2+x)^3, & -3 < x \leq -1 \\ x^{2/3}, & -1 < x < 2 \end{cases}$$
 is [IIT 2008]
 A) 0 B) 1 C) 2 D) 3
27. Let the function $g : (-\infty, \infty) \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be given by $g(u) = 2 \tan^{-1}(e^u) - \frac{\pi}{2}$. Then, g is [IIT 2008]
 A) even and is strictly increasing in $(0, \infty)$
 B) odd and is strictly decreasing in $(-\infty, \infty)$
 C) odd and is strictly increasing in $(-\infty, \infty)$
 D) neither even nor odd, but is strictly increasing in $(-\infty, \infty)$
28. Let f be a non-negative function on the interval $[0, 1]$. If

$$\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t) dt, \quad 0 \leq x \leq 1, \text{ and } f(0) = 0,$$
 then [IIT 2009]
 A) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$
 B) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$
 C) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$

$$D) f\left(\frac{1}{2}\right) > \frac{1}{2} \text{ and } f\left(\frac{1}{3}\right) < \frac{1}{3}$$

29. Interval in which $\tan^{-1}(\sin x + \cos x)$ is increasing (2009)
 A) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ B) $\left(0, \frac{\pi}{2}\right)$ C) $\left(\frac{\pi}{8}, \frac{5\pi}{4}\right)$ D) $\left(0, \frac{\pi}{8}\right)$
30. The second degree polynomial $f(x)$, satisfying $f(0) = 0$, $f(1) = 1$, $f'(x) > 0$ for all $x \in (0, 1)$
 A) $f(x) = \phi$
 B) $f(x) = ax + (1-a)x^2$; $\forall a \in (0, \infty)$
 C) $f(x) = ax + (1-a)x^2$, $a \in (0, 2)$
 D) no such polynomial
31. Let f, g and h be real valued functions defined on the interval $[0, 1]$ by

$$f(x) = e^{x^2} + e^{-x^2}, \quad g(x) = x \cdot e^{x^2} + e^{-x^2},$$

$$h(x) = x^2 \cdot e^{x^2} + e^{-x^2}.$$
 If a, b and c denotes respectively, the absolute maximum of f, g and h on $[0, 1]$ then [IIT 2010]
 A) $a = b$ and $c \neq b$ B) $a = c$ and $a \neq b$
 C) $a \neq b$ and $c \neq b$ D) $a = b = c$
32. Consider the polynomial

$$f(x) = 1 + 2x + 3x^2 + 4x^3.$$
 Let s be the sum of all distinct real roots of $f(x)$ and let $t = |s|$. The function $f'(x)$ is
 A) Increasing in $\left(-t, -\frac{1}{4}\right)$ and decreasing in $\left(-\frac{1}{4}, t\right)$
 B) Decreasing in $\left(-t, -\frac{1}{4}\right)$ and increasing in $\left(-\frac{1}{4}, t\right)$
 C) Increasing in $(-t, t)$
 D) Decreasing in $(-t, t)$
33. The radius of a right circular cylinder increases at the rate of 0.1 cm/min, and the

height decreases at the rate of 0.2 cm/min. The rate of change of the volume of the cylinder, in cm^3/min , when the radius is 2 cm and the height is 3 cm, is

- A) -2π B) $-\frac{8\pi}{5}$ C) $-\frac{3\pi}{5}$ D) $\frac{2\pi}{5}$

MULTI ANSWER QUESTIONS

34. Let $f(x) = |x^2 - 3x - 4|$, $-1 \leq x \leq 4$. Then

- A) $f(x)$ is monotonically increasing in $[-1, 3/2]$
 B) $f(x)$ is monotonically decreasing in $(3/2, 4]$
 C) the maximum value of $f(x)$ is $25/4$
 D) the minimum value of $f(x)$ is 0.

35. The critical point(s) of $f(x) = \frac{|2-x|}{x^2}$ is/are

- A) $x = 0$ B) $x = 2$ C) $x = 4$ D) $x = 1$

36. If the tangent at any point $P(4m^2, 8m^3)$ of

$x^3 - y^2 = 0$ is also a normal to the curve

$x^3 - y^2 = 0$, then the value of m is

- A) $m = \frac{\sqrt{2}}{3}$ B) $m = -\frac{\sqrt{2}}{3}$
 C) $m = \frac{3}{\sqrt{2}}$ D) $m = -\frac{3}{\sqrt{2}}$

37. The angle between the tangents at any point P and the line joining P to the origin, where P is a point on the curve in

$(x^2 + y^2) = c \tan^{-1} \frac{y}{x}$, c is a constant, is

- A) independent of x B) independent of y
 C) independent of x but dependent on y
 D) independent of y but dependent on x

38. Let $f(x) = \frac{1}{1+|x|} + \frac{1}{1+|x-1|}$ then

- A) $f(x)$ has global maximum
 B) $f(x)$ has local minimum
 C) $f(x)$ has absolute minimum
 D) $f(x)$ has local maximum

39. If $f(x) = (\sin^2 x - 1)^n$ then $x = \frac{\pi}{2}$ is a point of

- A) local maximum, if n is odd
 B) local minimum, if n is odd
 C) local maximum, if n is even
 D) local minimum, if n is even

40. Let $f(x) = \log(2x - x^2) + \sin \frac{\pi x}{2}$. Then which of the following is/are true?

- A) graph of f is symmetrical about the line $x = 1$
 B) maximum value of f is 1
 C) absolute minimum value of f does not exist
 D) $f(x)$ is a periodic function

41. The function $f(x) = x^{1/3} \cdot (x - 1)$

- A) has two inflection points
 B) has one point extremum
 C) is non-differentiable at $x = 0$
 D) Range of $f(x)$ is $[-3 \times 2^{-8/3}, \infty)$

42. The value of x for which the function

$f(x) = \int_0^x (1-t^2)e^{-t^2/2} dt$ has an extremum is

- A) 0 B) 1 C) -1 D) 2

43. A tangent to the curve $y = \int_0^x |t| dt$, which is parallel to the line $y = x$, cuts off an intercept from the y -axis equals to

- A) 1 B) $-1/2$ C) $1/2$ D) -1

44. If $f(x)$ and $g(x)$ are two positive and increasing functions, then

- A) $(f(x))^{g(x)}$ is always increasing
 B) If $(f(x))^{g(x)}$ is decreasing then $f(x) < 1$
 C) if $(f(x))^{g(x)}$ is increasing then $f(x) > 1$
 D) If $f(x) > 1$, then $(f(x))^{g(x)}$ is increasing

45. $f(x) = \begin{cases} |x+1| & ; -2 < x < 0 \\ 2 & ; x = 0 \\ \sqrt[3]{1-x} & ; 0 < x < 1 \\ \sqrt{x+1} & ; x \geq 1 \end{cases}$ Then $f(x)$

- A) has neither maximum nor minimum at $x = 0$
 B) has maximum at $x = 0$
 C) has neither maximum nor minimum at $x = 1$
 D) no global maximum

46. If $f(x) = \frac{x^2 - 1}{x^2 + 1}$, for every real number x ,

then the minimum value of f [IIT 1998]

- A) does not exist because f is unbounded
 B) is not attained even though f is bounded
 C) is equal to 1 D) is equal to -1

47. The number of values of x where the function $f(x) = \cos x + \cos(\sqrt{2}x)$ attains its maximum is [IIT 1998]
 A) 0 B) 1 C) 2 D) infinite
48. $f(x)$ is cubic polynomial with $f(2) = 18$ and $f(1) = -1$. Also $f(x)$ has local maxima at $x = -1$ and $f'(x)$ has local minima at $x = 0$, then [IIT - 2006]
 A) the distance between $(-1, 2)$, and $(a, f(a))$, where $x = a$ is the point of local minima is $2\sqrt{5}$
 B) $f(x)$ is increasing for $x \in [1, 2\sqrt{5}]$
 C) $f(x)$ has local minima at $x = 1$
 D) the value of $f(0) = 15$
49. Let

$$f(x) = \begin{cases} e^x, & 0 \leq x \leq 1 \\ 2 - e^{x-1}, & 1 < x \leq 2 \\ x - e, & 2 < x \leq 3 \end{cases}$$

$$g(x) = \int_0^x f(t) dt, x \in [1, 3]$$

then $g(x)$ has [IIT 2006]

- A) local maxima at $x = 1 + \ln 2$ and local minima at $x = e$
 B) local maxima at $x = 1$ and local minima at $x = 2$
 C) no local maxima D) no local minima
50. For function $f(x)$ $f(x) = x \cos \frac{1}{x}$, $x \geq 1$, (2009)
 A) for atleast one x in interval $[1, \infty)$, $f(x+2) - f(x) < 2$
 B) $\lim_{x \rightarrow 0} f'(x) = 1$
 C) $[1, \infty)$, $f(x+2) - f(x) > 2$
 D) $f'(x)$ is strictly decreasing in the interval $[1, \infty)$
51. If $f(x) = \int_0^x e^{t^2} (t-2)(t-3) dt$ for all $x \in (0, \infty)$, then
 A) f has a local maximum at $x=2$
 B) f is decreasing on $(2, 3)$

C) there exists some $c \in (0, \infty)$ such that

$$f''(c) = 0$$

D) f has a local minimum at $x=3$

ASSERTION & REASON QUESTIONS

52. Statment-1: Both $\sin x$ and $\cos x$ are

decreasing functions in the interval $\left(\frac{\pi}{2}, \pi\right)$

Statement-2: If a differentiable function decreases in an interval (a, b) , then its derivative also decreasing in (a, b) .

which of the following is true? [IIT 2000]

- A) both st1 and st2 are wrong
 B) both st1 and st2 are correct but st2 is not correct explanation for st1
 C) both st1 and st2 are correct and st2 is correct explanation for st1
 D) st1 is correct and st2 is wrong

INTEGER QUESTIONS

53. Equation of the normal to the curve

$y = (1+x)^y + \sin^{-1}(\sin^2 x)$ at $x = 0$ is $x + y = k$, then k is

54. If the acute angles between the curves

$y = |x^2 - 1|$ and $y = |x^2 - 3|$ at their points of

intersection be θ such that $\tan \theta = \frac{4\sqrt{2}}{k}$

then k is equal to

55. A right triangle is drawn in a semi circle of

radius $\frac{1}{2}$ with one of its legs along the

diameter. If the maximum area of the triangle is M , then the value of $32\sqrt{3}M$ is

56. The maximum value of the function $f(x) = 2x^3 - 15x^2 + 36x - 48$ on the set

$A = \{x | x^2 + 20 \leq 9x\}$ is (2009)

57. If the function $f(x) = x^3 + e^{x/2}$ and

$g(x) = f^{-1}(x)$, then the value of $g'(1)$ is (2009)

58. Let $f(x)$ be a function defined on $\mathbb{R}$ such

that $f^4(x) = 2010(x-2009)(x-2010)^2(x-2011)^3(x-2012)^4$ for all $x \in \mathbb{R}$. If g is a function defined on $\mathbb{R}$ with values in $(0, \infty)$ such that

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$f(x) = \ln(g(x))$ for all $x \in \mathbb{R}$. The number of the points in $\mathbb{R}$ at which g has a local maximum is

59. Let $f(x)$ be a real valued differentiable function on $\mathbb{R}$ such that $f(1) = 1$. If y-intercept of the tangent at any point $p(x, y)$ on the curve $y = f(x)$ is equals to the cube on the abscissa of p , then the value of $f(-3)$ is equals to.....
60. The number of distinct real roots of $x^4 - 4x^3 + 12x^2 + x - 1 = 0$ is (2011)
61. Let $p(x)$ be a real polynomial of least degree which has a local maximum at $x=1$ and a local minimum at $x=3$. If $p(1) = 6$ and $p(3) = 2$, then $p'(0)$ is

COMPREHENSION QUESTIONS

PASSAGE - 1

FOR PROBLEMS - 62-64

Consider the curve $x = 1 - 3t^2$, $y = t - 3t^3$.

If a tangent at point $(1 - 3t^2, t - 3t^3)$ inclined at an angle θ to the positive x -axis and another tangent at point $P(-2, 2)$ cuts the curve again at Q .

62. The value of $\tan \theta + \sec \theta$ is equal to
A) $3t$ B) t C) $t - t^2$ D) $t^2 - 2t$
63. The point Q will be
A) $(1, -2)$ B) $\left(-\frac{1}{3}, -\frac{2}{9}\right)$ C) $(-2, 1)$ D) $(1, 0)$
64. The angle between the tangents at P and Q will be
(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{3}$

PASSAGE - 2

FOR PROBLEMS - 65-66

Let $f'(\sin x) < 0$ and $f''(\sin x) > 0 \forall x \in \left(0, \frac{\pi}{2}\right)$

and $g(x) = f(\sin x) + f(\cos x)$.

65. Which of the following is true in $\left(0, \frac{\pi}{2}\right)$?
A) g' is increasing B) g' is decreasing

C) g' has a point of minima

D) g' has a point of maxima

66. Which of the following is true?

A) $g(x)$ is decreasing in $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$

B) $g(x)$ increasing in $\left[0, \frac{\pi}{4}\right]$

C) $g(x)$ is monotonically increasing

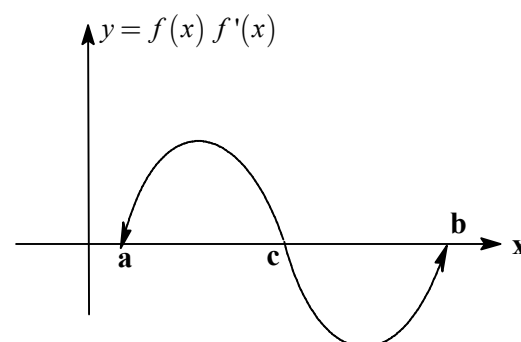
D) None of these

PASSAGE : 3

FOR PROBLEMS - 67 - 69

$f(x)$ is continuous and differentiable function. Given, $f(x)$ assumes values of the form $\pm\sqrt{I}$ where I denotes set of whole numbers whenever $x = a$ or b ; otherwise $f(x)$ assumes real values. Also

$$f(c) = -\frac{3}{2} \text{ and } |f(a)| \leq |f(b)|.$$



67. The number of rational values that $f(a) + f(b) + f(c)$ can assume is / are
A) 4 B) 2 C) 3 D) 5
68. The number of values that $(f(a))^2 + (f(b))^2 + (f(c))^2$ can assume is
A) 4 B) 2 C) 3 D) 7
69. The possible number of triplets $(f(a), f(b), f(c))$ is / are
A) 4 B) 5 C) 7 D) 6

PASSAGE-4

FOR PROBLEMS - 70 - 71

Let $f(x) = (1-x)^2 \sin^2 x + x^2$ for all $x \in \mathbb{R}$,

and let $g(x) = \int_1^x \left(\frac{2(t-1)}{t+1} - \ln t \right) f(t) dt$ for

all $x \in (1, \infty)$

70. Which of the following is true?

A) g is increasing on $(1, \infty)$

B) g is decreasing on $(1, \infty)$

C) g is increasing on $(1, 2)$ and decreasing on $(2, \infty)$

D) g is decreasing on $(1, 2)$ and increasing on $(2, \infty)$

71. Consider the statements:

P: There exists some $x \in \mathbf{R}$ such that

$$f(x) + 2x = 2(1 + x^2)$$

Q: There exists some $x \in \mathbf{R}$ such that

$$2f(x) + 1 = 2x(1 + x)$$

A) both P and Q are true

B) P is true and Q is false

C) P is false and Q is true

D) both P and Q are false

PASSGE-5 :- (72-74)

The relation between the distance of any point on the curve from the origin & length of the perpendicular from the origin to the tangent at the point is called pedal equation of the curve equation of tangent at a point (h, k) , $y - k = f'(h)(x - h)$

p = length of perpendicular from $(0, 0)$ to

$$\text{the tangent} = \frac{|k - hf'(h)|}{\sqrt{1 + (f')^2}}$$

$$k = f(h), r^2 = h^2 + k^2$$

P-perpendicular distance $(0, 0)$ to the

tangent at (h, k) r – distance from origin with the above information answer the following questions :-

72. The pedal equation of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\text{A) } \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} - \frac{r^2}{a^2 b^2}$$

$$\text{B) } \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{r^2}{a^2 b^2}$$

$$\text{C) } \frac{2}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} - \frac{r^2}{a^2 b^2}$$

$$\text{D) } \frac{2}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{r^2}{a^2 b^2}$$

73. The equation of a curve is given in parametric form

$x = a \cos^3 \theta, y = a \sin^3 \theta$, the pedal equation of this curve is

$$\text{A) } r^2 = a^2 + 3p^2 \quad \text{B) } r^2 = a^2 - 3p^2$$

$$\text{C) } r^2 = 3a^2 - p^2 \quad \text{D) } r^2 = 3a^2 + p^2$$

74. The cartesian equation of curve is given

by $c^2(x^2 + y^2) = x^2 y^2$. the pedal equation of this curve is

$$\text{A) } \frac{1}{p^2} + \frac{3}{r^2} = \frac{1}{c^2}$$

$$\text{B) } \frac{3}{p^2} + \frac{1}{r^2} = \frac{1}{c^2}$$

$$\text{C) } \frac{1}{p^2} - \frac{3}{r^2} = \frac{1}{c^2}$$

$$\text{D) } \frac{3}{p^2} - \frac{1}{r^2} = \frac{1}{c^2}$$

PASSGE-6 :- (75-77)

Tangent at any point P_1 (other than $(0,0)$) on the curve $y = x^3$ meets the curve again at P_2 . Tangent at P_2 meets curve again at P_3 and so on.

Choose correct answer

75. Abscissae of $P_1, P_2, P_3, \dots, P_n$ are in
A) A.P B) G.P C) H.P D) None
76. Ordinates of $P_1, P_2, P_3, \dots$ are in
A) A.P B) G.P C) H.P D) None
77. Ratio of areas of triangles $P_2P_3P_4$ and $P_1P_2P_3$ is
A) 4 B) 8 C) 16 D) 32

MATRIX-MATCHING QUESTIONS

78. Let $f(x)$ be a real valued function

defined by $f(x) = x^2 - 2|x|$ and

$$g(x) = \begin{cases} \text{minimum}\{f(t); -2 \leq t \leq x\}, & x \in [-2, 0] \\ \text{maximum}\{f(t); 0 \leq t \leq x\}, & x \in [0, 3] \end{cases}$$

Column-I

- A) $f(x)$ is not continuous at x equal to
B) $g(x)$ is not derivable at x equal to
C) The points of local extremum of $g(x)$ is/are
D) Absolute maximum value of $g(x)$ is equal to

Column-II

- (P) -2
(Q) 0
(R) 1
(S) 2
(T) 3

SUBJECTIVE QUESTIONS

79. Show that $2\sin x + 2\tan x \geq 3x$ where $0 \leq x < \pi/2$. (1990-4M)

80. Let $f(x) =$

$$\begin{cases} -x^3 + \frac{(b^3 - b^2 + b - 1)}{(b^2 + 3b + 2)}, & 0 \leq x < 1 \\ 2x - 3, & 1 \leq x \leq 3 \end{cases}$$

Find all possible real values of b such that $f(x)$ has the smallest value at $x = 1$.

(1993-5M)

81. The curve $y = ax^3 + bx^2 + cx + 5$, touches the x -axis at $P(-2, 0)$ and cuts the y -axis at a point Q , where its gradient is 3. Find a, b, c . (1994-5M)

82. Let $f(x) = \begin{cases} xe^{ax}, & x \leq 0 \\ x + ax^2 - x^3, & x > 0 \end{cases}$. Where a is a positive constant. Find the interval in which $f'(x)$ is increasing. (1996-3M)

83. Using the relation $2(1 - \cos x) < x^2$, $x \neq 0$ or otherwise, prove that $\sin(\tan x) \leq x$,

$$\forall x \in \left[0, \frac{\pi}{4}\right]. \quad (2003-4M)$$

**KEY
LEVEL-VI****KEY****SINGLE ANSWER QUESTIONS**

- | | | | |
|------|------|------|------|
| 1)C | 2)A | 3)A | 4)C |
| 5)A | 6)A | 7)B | 8)A |
| 9)B | 10)A | 11)B | 12)A |
| 13)D | 14)D | 15)C | 16)D |
| 17)B | 18)A | 19)B | 20)A |
| 21)B | 22)A | 23)A | 24)C |
| 25)B | 26)C | 27)C | 28)C |
| 29)D | 30)C | 31)D | 32)B |
| 33)D | | | |

MULTI ANSWER QUESTIONS

- | | |
|----------------|----------------|
| 34) A, B, C, D | 35) B, C |
| 36) A, B | 37) A, B |
| 38) A, B, D | 39) A, D |
| 40) A, B, C | 41) A, B, C, D |
| 42) B, C | 43) B, C |
| 44) C, D | 45) B, C, D |
| 46) D | 47) B |
| 48) B, C | |
| 49) A, B | 50) B, C, D |
| 51) A, B, C, D | |

ASSERTION & REASON QUESTIONS

52) D

INTEGER QUESTION

53) 1 54) 7 55) 9 56) 7

57) 2 58) 1 59) 9 60) 2

61) 9

COMPERHENSION QUESTIONS

62) A 63) B 64) C 65) A

66) D 67) C 68) A 69) D

70) B 71) C 72) A 73) B

74) A 75) B 76) D 77) C

MATRIX- MATCHING QUESTIONS78) $A \rightarrow Q, B \rightarrow Q, S, C \rightarrow Q, R, S, D \rightarrow T$ **SUBJECTIVE QUESTIONS**

81. $b \in (-2, -1) \cup (1, \infty)$

82. $a = -\frac{1}{2}, b = -\frac{3}{4}, c = 3$

83. $\left(\frac{-2}{a}, \frac{a}{3}\right)$

LEVEL VI**HINTS****SINGLE ANSWER QUESTIONS**

- Let $g(x) = f^3(x) \Rightarrow g'(x) = 3f^2(x) \cdot f'(x)$
 $\therefore f: [2, 7] \rightarrow [0, \infty) \Rightarrow g: [2, 7] \rightarrow [0, \infty)$
 Using Legrange's mean value theorem on $g(x)$, we get $g'(c) = \frac{g(7) - g(2)}{5}, c \in (2, 7)$
 $\Rightarrow 3f^2(c) \cdot f'(c) = \frac{f^3(7) - f^3(2)}{5}$
 $5f^2(c) \cdot f'(c) = \frac{(f(7) - f(2))(f^2(7) + f^2(2) + f(7) \cdot f(2))}{3}$
 Hence, (c) is the correct answer.
- Correct option can be identified by sketching graph and writing equation of tangent and line.
- $|f(1) + 3| < |f(1)| + 3 \Rightarrow f(1) < 0$
 $|f(2) + 10| = |f(2)| + 10 \Rightarrow f(2) > 0$
- $f''(x) \geq 0 \Rightarrow f'(x)$ is increasing function &

APPLICATION OF DERIVATIVES

$$g'(x) = (2x-2) \left[f'(x^2-2x+8) - f'(14+2x-x^2) \right]$$

For $g'(x) > 0$

(i) $x > 1$ and $x^2 - 2x + 8 > 14 + 2x - x^2$

or

(ii) $x < 1$ and $x^2 - 2x + 8 < 14 + 2x - x^2$

5. $3 \sin x - 4 \sin^3 x = \sin 3x$ which increases for

$$3x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \Rightarrow x \in \left(-\frac{\pi}{6}, \frac{\pi}{6}\right) \text{ whose length}$$

is $\frac{\pi}{3}$.

6. $f'\left(\frac{1-}{2}\right) = -1$ and

$$f'(1/2^+) = -2\left(\frac{1}{2} - \frac{1}{2}\right) = 0$$

$$\therefore f'\left(\frac{1-}{2}\right) \neq f'(1/2^+)$$

$$\therefore f \text{ is not differentiable at } \frac{1}{2} \in (0, 1)$$

7. Equation of tangent to the ellipse

$$\frac{\sqrt{3}x \cos \theta}{9} + y \sin \theta = 1$$

$$\therefore \text{Sum of intercepts} =$$

$$S = 3\sqrt{3} \sec \theta + \csc \theta$$

For min. value of $S, \frac{dS}{d\theta} = 0$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{3}} = \tan \frac{\pi}{6} \Rightarrow \theta = \frac{\pi}{6}$$

8. $\frac{dy}{dx} = -\frac{x_1^2}{y_1^2}$

tangent $y - y_1 = -\frac{x_1^2}{y_1^2}(x - x_1)$ (1)

$x_2^3 + y_2^3 = a^3$ (2)

$x_1^3 + y_1^3 = a^3$ (3)

APPLICATION OF DERIVATIVES

putting $y = y_2$ and $x = x_2$ and solving with (2) and (3) we get result.

9. Consider the function $f(x) = \tan x$, defined on $[a, b]$ such that $a, b \in \left(0, \frac{\pi}{2}\right)$.

Applying Lagrange's mean value theorem, we have

$$f'(c) = \frac{f(b) - f(a)}{b - a} \text{ for some } c \in (a, b)$$

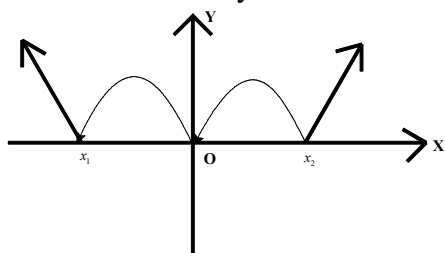
$$\Rightarrow \sec^2 C = \frac{\tan b - \tan a}{b - a} \Rightarrow f(a, b) = \sec^2 C$$

$$\Rightarrow f(a, b) > 1 \quad [\because \sec^2 C > 1 \text{ as } C \in (0, \pi/2)]$$

10. $f(x) = 0$ has roots 1, 2, 3 only
 $\Rightarrow$ Any one of 1, 2, 3, is a repeated root of $f(x) = 0 \Rightarrow f'(1)$ or $f'(2)$ or $f'(3)$ any one of them must be zero.
Hence, (a) is the correct answer.

11. Given that $f(|x|) = 0$ has 8 real roots.
 $\Rightarrow f(x) = 0$ has 4 positive roots.
Since $f(x)$ is a polynomial of degree 5, $f(x)$ cannot have even number of real roots.
 $\Rightarrow f(x)$ has all the five roots real, in which four positive and one root is negative.
Hence, (b) is the correct answer.

12. Here, $f(x) = |x^2 + a|x + b|$ has three points of non-differentiability.



$\therefore f(x)$ is non-differentiable at $x = 0, x_1, x_2$
 $\Rightarrow x^2 + ax + b = 0$ has one root zero and other positive root.
 $\Rightarrow b = 0$ and $a < 0$, is one of the case.
Hence, (a) is the correct answer.

13. $|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$
 $\Rightarrow \frac{d}{dx}|f(x)| = \begin{cases} f'(x) & f(x) > 0 \\ -f'(x) & f(x) < 0 \end{cases}$
Now as $f(x)$ and $f'(x)$ keep opposite sign,

then $\frac{d}{dx}|f(x)| < 0$.

Hence: $|f|$ is decreasing

14. $f'(x) = 12x^2 - 2x - 2 = 2[6x^2 - x - 1] = 2(3x+1)(2x-1)$

$$g(x) = \begin{cases} f(x) & \text{if } 0 \leq x < \frac{1}{2} \\ f\left(\frac{1}{2}\right) & \text{if } \frac{1}{2} \leq x \leq 1 \\ 3-x & \text{if } 1 < x \leq 2 \end{cases}$$

Hence

$$\Rightarrow g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right) = f\left(\frac{1}{4}\right) + f\left(\frac{1}{2}\right) + g\left(\frac{5}{4}\right) = \frac{5}{2}$$

15. Consider the function $f(x) = \frac{x^2}{(x^3 + 200)}$

$$f'(x) = x \frac{(400 - x^3)}{(x^3 + 200)^2} = 0$$

When $x = (400)^{1/3}$

$$x = (400)^{1/3} - h \Rightarrow f'(x) > 0$$

$$x = (400)^{1/3} + h \Rightarrow f'(x) < 0 \therefore f(x) \text{ has maxima at } x = (400)^{1/3}$$

Since $7 < (400)^{1/3} < 8$, either a_7 or a_8 is the greatest term of the sequence.

$$\therefore a_7 = \frac{49}{543} \text{ and } a_8 = \frac{8}{89} \text{ and } \frac{49}{543} > \frac{8}{89}$$

$$\Rightarrow a_7 = \frac{49}{543} \text{ is the greatest term.}$$

16. Let there be a value of k for which $x^3 - 3x + k = 0$ has two distinct roots between 0 and 1.

Let a, b be two distinct roots of

$$x^3 - 3x + k = 0 \text{ lying between 0 and 1 such that } a < b. \text{ Let}$$

$$f(x) = x^3 - 3x + k. \text{ Then } f(a) = f(b) = 0.$$

Since between any two roots of a polynomial $f(x)$, there exists at least one root of its derivative $f'(x)$. Therefore,

$$f'(x) = 3x^2 - 3 \text{ has at least one root between } a \text{ and } b.$$

But $f'(x) = 0$ has two roots equal to ± 1 which do not lie between a, b .

Hence: $f(x) = 0$ has no real roots lying between 0 and 1 for any value of k .

17. Here, $\left(\sqrt{-3+4x-x^2}+4\right)^2+(x-5)^2$, represents the square of the distance between circle $y=\sqrt{-3+4x-x^2}$ and point $(5, -4)$ ie, Maximum distance between $x^2+y^2-4x+3=0$ and $(5, -4)$ squared

$$\Rightarrow PQ^2 = (PC + \text{radius})^2 = \left(\sqrt{(5-2)^2 + (4-0)^2} + 1\right)^2 = 6^2 = 36$$

Hence, (b) is the correct answer.

18. $f(x) = \int_{5\pi/4}^x (3\sin t + 4\cos t) dt, x \in \left[\frac{5\pi}{4}, \frac{4\pi}{3}\right]$

$$f'(x) = 3\sin x + 4\cos x$$

$f'(x) < 0$ as $\sin x, \cos x$ are negative for

$$x \in \left[\frac{5\pi}{4}, \frac{4\pi}{3}\right] \Rightarrow f(x)\bigg|_{\min} = f\left(\frac{4\pi}{3}\right)$$

$$= \int_{5\pi/4}^{4\pi/3} (3\sin t + 4\cos t) dt = \frac{3}{2} + \frac{1}{\sqrt{2}} - 2\sqrt{3}$$

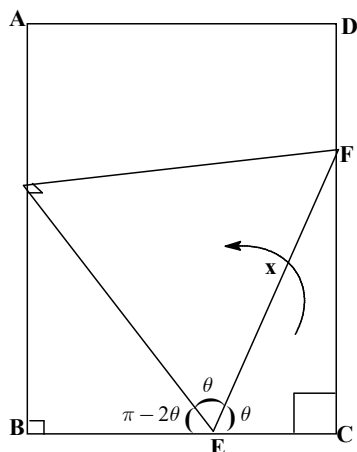
Hence, (a) is the correct answer.

19. $f'(x) = \frac{(e+x)\ln(e+x) - (\pi+x)\ln(\pi+x)}{(e+x)(\pi+x)(\ln(e+x))^2}$

$0 < \text{on } (0, \infty)$ since $1 < e < \pi$

$\therefore f(x)$ is decreases on $(0, \infty)$.

20. Let ABCD be the rectangular sheet whose corner C is folded over along EF so as to reach the edge AB at C'.



Let $EF = x$

$$\therefore \angle FEC = \theta = \angle FEC'$$

APPLICATION OF DERIVATIVES

From $\triangle BEC'$ we have $BE = C'E \cos(\pi - 2\theta)$

$$\Rightarrow BE = -x \cos \theta \cdot \cos 2\theta$$

$$\therefore BC = BE + EC$$

$$1 = -x \cos \theta \cos 2\theta + x \cos \theta$$

$$\Rightarrow x = \frac{1}{\cos(1 - \cos 2\theta)} \dots\dots\dots (1)$$

x to be minimum, Z has to be maximum

i, e $Z = \frac{1}{x} = \cos \theta (1 - \cos 2\theta) \dots\dots\dots (2)$

Differentiating Eq. (2) w.r.t. θ , we get

$$\frac{dZ}{d\theta} = \cos \theta (+2 \sin 2\theta) - \sin \theta (1 - \cos 2\theta)$$

and

$$\frac{d^2Z}{d\theta^2} = \cos \theta (4 \sin 4\theta) - 2 \sin 2\theta \cdot \sin \theta - \sin \theta$$

$$(2 \sin 2\theta) - \cos \theta (1 - \cos 2\theta)$$

For maximum/minimum,

$$\frac{dZ}{d\theta} = 0 \Rightarrow 2 \sin \theta (2 - 3 \sin^2 \theta) = 0$$

$$\therefore \sin \theta = +\sqrt{2/3}$$

($\because \sin \theta \neq 0$)

When

$$\sin \theta = \sqrt{\frac{2}{3}} \Rightarrow \frac{d^2Z}{d\theta^2} = -\frac{5}{5\sqrt{3}} - \frac{16}{3\sqrt{3}} - \frac{1}{\sqrt{3}} = \frac{8}{\sqrt{3}} < 0$$

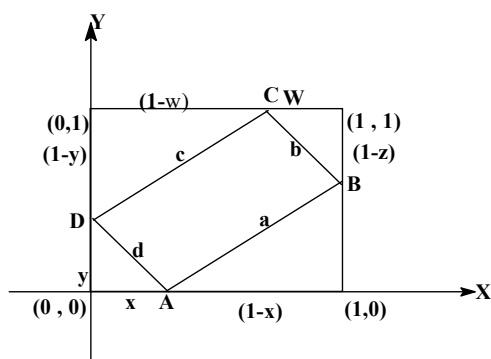
Hence: Z is maximum.

$$\Rightarrow x = \frac{1}{Z} \text{ is minimum}$$

[from Eq.(1)] $\therefore x$ is minimum.

$$\Rightarrow \min x \Rightarrow \frac{1}{Z} = \frac{1}{(1 + \sqrt{3})\left(1 + \frac{1}{3}\right)} = \frac{3\sqrt{3}}{4} \text{ unit}$$

21. Let S be the square of unit area and ABCD be the quadrilateral of sides a, b, c , and d .



Here

$$a^2 = (1-x)^2 + z^2, b^2 = w^2 + (1-z)^2$$

$$c^2 = (1-w)^2 + (1-y)^2, d^2 = x^2 + y^2$$

Adding all the above, $a^2 + b^2 + c^2 + d^2$

$$= \{x^2 + (1-x)^2\} + \{y^2 + (1-y)^2\} + \{z^2 + (1-z)^2\} + \{w^2 + (1-w)^2\}$$

where $0 \leq x, y, z, w \leq 1$

Let us consider function,

$$f(x) = x^2 + (1-x)^2, 0 \leq x \leq 1$$

Then $f'(x) = 2x - 2(1-x)$

Let $f'(x) = 0$ for maximum/minimum,

$$\Rightarrow 4x - 2 = 0 \Rightarrow x = 1/2$$

Again, $f''(x) = 4 > 0$ when $x = 1/2$

$\therefore f(x)$ is minimum at $x = 1/2$ and maximum

$$\text{at } x = 1 \Rightarrow 2 \leq a^2 + b^2 + c^2 + d^2 \leq 4$$

22. $\frac{dy}{dx} = +1$

23. $f(x) = x^3 + bx^2 + cx + d, 0 < b^2 < c$

$$f'(x) = 3x^2 + 2bx + c$$

$$\text{Discriminant} = 4b^2 - 12c = 4(b^2 - 3c) < 0$$

$$\therefore f'(x) > 0 \quad \forall x \in \mathbb{R} \Rightarrow f(x) \text{ is strictly}$$

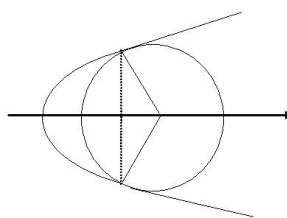
$$\forall x \in \mathbb{R}$$

24. $\lim_{x \rightarrow 0^+} f(x) = f(0) \Rightarrow$

$$\lim_{x \rightarrow 0^+} x^\alpha \log x = 0 \Rightarrow \lim_{x \rightarrow 0^+} \frac{\log x}{1/x^\alpha} = 0$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \frac{1/x}{(-\alpha)x^{-\alpha-1}} = 0 \Rightarrow \lim_{x \rightarrow 0^+} \frac{x^\alpha}{-\alpha} = 0 \Rightarrow \alpha > 0$$

25. $(x-3)^2 + (y-0)^2 = (2\sqrt{2})^2$

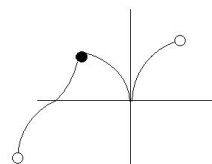


$$x^2 + 4x - 6x + = 0$$

$$(x-1)^2 = 0, x = 1, 1$$

Ans. touch each other exactly two points.

26. $f(x) = (2+x)^3, -3 < x < -1 = x^{2/3},$
 $-1 < x < 2$



The total number of local maximum or minimum = 2.

27. $g(u) = 2 \tan^{-1}(e^u) - \frac{\pi}{2}$

$$g'(u) = \frac{2e^u}{1+e^{2u}} > 0$$

Hence $g(u)$ is increasing function

$$g(u) = \tan^{-1}(e^u) - \cot^{-1}(e^u)$$

$$g(-u) = \tan^{-1}(e^{-u}) - \cot^{-1}(e^{-u})$$

$$= \tan^{-1}\left(\frac{1}{e^u}\right) - \cot^{-1}\left(\frac{1}{e^u}\right) = \cot^{-1}(e^u) - \tan^{-1}(e^u)$$

$$= -g(u)$$

Hence $g(u)$ is odd strictly increasing in $(-\infty, \infty)$

28. $f' = \pm \sqrt{1-f^2} \Rightarrow f(x) = \sin x$ or
 $f(x) = -\sin x$ (not possible) $\Rightarrow f(x) = \sin x$

Also, $x > \sin x \quad \forall x > 0$

29. $f'(x) > 0$ if $\cos x > \sin x$

30. Let $f(x) = bx^2 + ax + c$

$$\text{as } f(0) = 0 \Rightarrow c = 0$$

$$f(1) = 1 \Rightarrow a + b = 1$$

$$\therefore f(x) = ax + (1-a)x^2$$

$$\text{as } f'(x) > 0, \text{ for } x \in (0, 1)$$

$$\Rightarrow f'(x) = a + 2(1-a)x > 0 \text{ for } x \in (0, 1)$$

$$\text{when } x=0 \Rightarrow a > 0$$

$$\text{when } x=1 \Rightarrow a + 2 - 2a > 0 \Rightarrow a - 2 < 0$$

$$\therefore a \in (0, 2)$$

31. $f^1(x) = 2xe^{x^2} - 2xe^{-x^2} = 2x[e^{x^2} - e^{-x^2}] \geq 0 \forall x \in [0, 1]$
 $g^1(x) = e^{x^2} + 2x^2 \cdot e^{x^2} - 2x \cdot e^{-x^2} > 0 \forall x \in [0, 1]$
 $h^1(x) > 0 \forall x \in [0, 1]$ hence f, g, h are
 increasing functions in $[0, 1]$

Maximum of $f = f(1)$,

and that of $g(x)$ and $h(x)$ and $g(1)$ and $h(1)$, Hence

$$f(1) = g(1) = h(1) = e + \frac{1}{e} \Rightarrow a = b = c$$

32. $f'(x) = 2 + 6x + 12x^2 > 0 \forall x \in R$.

Hence $f(x)$ has only one real root belongs to

$$\left(-\frac{3}{4}, -\frac{1}{2}\right) \text{ since } f\left(-\frac{1}{2}\right)f\left(-\frac{3}{4}\right) < 0$$

$$\text{hence } s \in \left(-\frac{3}{4}, -\frac{1}{2}\right)$$

$$f^{11}(x) = 6 + 24x = 0 \Rightarrow x = -\frac{1}{4}$$

$$f^{11}(x) < 0 \text{ for } x < -\frac{1}{4} \text{ and}$$

$$f^{11}(x) > 0 \text{ for } x > -\frac{1}{4}$$

$$f^1(x) \text{ decreasing in } \left(-t, -\frac{1}{4}\right)$$

$$\text{and increasing in } \left(\frac{1}{4}, t\right)$$

33. Given, $v = \pi r^2 h$

$$\frac{dV}{dt} = \pi \left(r^2 \frac{dh}{dt} + 2r \frac{dr}{dt} h \right) = \pi r \left(r \frac{dh}{dt} + 2h \frac{dr}{dt} \right)$$

$$\text{Also given, } \frac{dr}{dt} = \frac{1}{10} \text{ and } \frac{dh}{dt} = \frac{-2}{10}$$

$$\Rightarrow \frac{dV}{dt} = \pi r \left(r \left(\frac{-2}{10} \right) + 2h \left(\frac{1}{10} \right) \right) = \frac{\pi r}{5} (-r + h)$$

when $r = 2$ and $h = 3$

$$\text{Then, } \Rightarrow \frac{dV}{dt} = \frac{2\pi}{5}$$

MULTI ANSWER QUESTIONS

34. $x^2 - 3x - 4 = (x - 4)(x + 1)$. So in $-1 \leq x \leq 4$,
 $x^2 - 3x - 4 < 0$
 $\therefore f(x) = -(x^2 - 3x - 4)(4 - x)(x + 1)$
 $\therefore f(x) = -(x + 1) + 4 - x = 3 - 2x$
 $\therefore$ in $-1 \leq x < \frac{3}{2}$, $f(x) > 0$ and in $\frac{3}{2} < x \leq 4$,
 $f'(x) < 0$

As $f(x)$ is continuous at $x = \frac{3}{2}$ we find $f\left(\frac{3}{2}\right)$ is maximum value.

The minimum value = the least among $\{f(-1), f(4)\}$

35. Obviously, at $x = 0$, $f(x) = \infty$

$\therefore f'(0)$ does not exist.

So, $x = 0$ is a critical point

$$\text{Now, } f(x) = \frac{2-x}{x^2}, 0 < x < 2, \frac{x-2}{x^2}, x \geq 2$$

At $x = 2, 4$ the function $f(x)$ is not differentiable. So, they are critical points.

36. $x^3 - y^2 = 0 \Rightarrow 2y \times \frac{dy}{dx} = 3x^2 \quad (1)$

Slope of the tangent at

$$P = \frac{dy}{dx} \Big|_P = \frac{3x^2}{2y} \Big|_{(4m^2, 8m^2)} = 3m$$

$\therefore$ Equation of the tangent at P is

$$y - 8m^3 = 3m(x - 4m^2) \text{ or } y = 3mx - 4m^2 \quad (2)$$

It cuts the curve again at point Q. Solving (1) and (2), we get $x = 4m^2, m^2$

Put $x = m^2$ in equation (2)

$$y - 3m(m^2) - 4m^3 = -m^3 \quad \therefore Q \text{ is } (m^2, -m^3)$$

Slope of the tangent at

$$Q = \frac{dy}{dx} \Big|_{(m^2, -m^3)} = \frac{3(m^4)}{2 \times (-m^3)} = \frac{-3}{2}m$$

$$\text{Slope of the normal at } Q = \frac{1}{(-3/2)m} = \frac{2}{3m}$$

APPLICATION OF DERIVATIVES

Since tangent at P is normal at Q

$$Q \Rightarrow \frac{2}{3m} = 3m \Rightarrow 9m^2 = 2$$

37. Let P (x, y) be a point on the curve in

$(x^2 + y^2) = c \tan^{-1} \frac{y}{x}$, Differentiating both sides with respect to x, we get

$$\frac{2x + 2yy'}{(x^2 + y^2)} = \frac{c(xy' - y)}{(x^2 + y^2)} \Rightarrow y' = \frac{2x + cy}{cx - 2y} = m_1$$

(say) Slope of OP = $\frac{y}{x} = m_2$ (say)

(where O is origin)

Let the angle between the tangents at P and OP be θ

$$\Rightarrow \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{\frac{2x + cy}{cx - 2y} - \frac{y}{x}}{1 + \frac{(2x + cy)y}{(cx - 2y)x}} \right| = \frac{2}{c}$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{2}{c} \right) \text{ which is independent of } x \text{ and } y.$$

38. Find $f'(x)$ and $f''(x)$

39. $f(x) = (\sin^2 x - 1)^n$

$$f\left(\frac{\pi}{2}\right) = 0$$

$$f\left(\frac{\pi^+}{2}\right) = (\rightarrow 0^-)^n \text{ and } f\left(\frac{\pi^-}{2}\right) = (\rightarrow 0^-)^n$$

If n is even $f\left(\frac{\pi^+}{2}\right)$ and $f\left(\frac{\pi^-}{2}\right) > 0$, then $x = \frac{\pi}{2}$ is the point of minima.

If n is odd $f\left(\frac{\pi^+}{2}\right)$ and $f\left(\frac{\pi^-}{2}\right) < 0$, then

$x = \frac{\pi}{2}$ is the point of maxima.

40. $f(x) = x^4 (12 \log_e x - 7); x > 0$

$$= \log(1 - (x-1)^2) + \sin \frac{\pi x}{2}$$

$$f(1-x) = \log(1 - (x-1)^2) + \sin \frac{\pi(1-x)}{2}$$

JEE ADVANCED - VOL - III

$$= \log(1 - x^2) + \cos \frac{\pi x}{2}$$

$$\text{Also } f(1+x) = \log(1 - (x-1)^2) + \sin \frac{\pi(1+x)}{2}$$

$$= \log(1 - x^2) + \cos \frac{\pi x}{2}$$

Hence, function is symmetrical about line $x = 1$

$$\text{Also } f(1) = 1$$

Also, for domain of the function is $2x - x^2 > 0$

or $x \in (0, 2)$

For $x > 1$, $f(x)$ decreases hence $x = 1$ is point of maxima.

Also, maxima value of the function is 1.

Also, $f(x) \rightarrow \infty$ when $x \rightarrow 2$, hence absolute minimum value of f does not exist.

41. $f(x) = x^{1/3} \cdot (x-1)$

$$\Rightarrow \frac{df(x)}{dx} = \frac{4}{3} x^{1/3} - \frac{1}{3} \cdot \frac{1}{x^{2/3}} = \frac{1}{3x^{2/3}} [4x - 1]$$

$\Rightarrow f'(x)$ changes sign from -ve to +ve, at $x = 1/4$, which is point of minima.

Also, $f'(x)$ does not exist at $x = 0$ as $f(x)$ has vertical tangent at $x = 0$.

$$f''(x) = \frac{4}{9} \cdot \frac{1}{x^{2/3}} + \frac{1}{3} \cdot \frac{2}{3} \cdot \frac{1}{x^{5/3}}$$

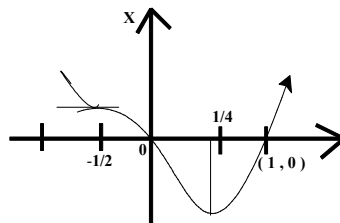
$$= \frac{2}{9x^{2/3}} \left[2 + \frac{1}{x} \right] = \frac{2}{9x^{2/3}} \left[\frac{2x+1}{x} \right]$$

$\therefore f''(x) = 0$ at $x = -\frac{1}{2}$ which is the point of

inflection at $x = 0$, $f''(x)$ does not exist but

$f''(x)$ changes sign, hence $x = 0$ is also the point of inflection.

From the above information the graph of $y = f(x)$ is as shown.



Also, minimum value of $f(x)$ is at $x = 1/4$ which is $-3 \times 2^{-8/3}$

Hence, range is $[-3 \times 2^{-8/3}, \infty)$

42. $f'(x) = (1-x^2)e^{-x^2/2}$

For extremum, $(1-x^2)e^{-x^2/2} = 0$, ie $x = 1, -1$.

43. Differentiating w.r.t. x , $\frac{dy}{dx} = |x| = 1$ because the slope of $y = x$ is 1

$\therefore$ at (α, β) , $\frac{dy}{dx}_{\alpha, \beta} = 1 = |\alpha| \therefore a = 1, -1$

$\therefore$ when $\alpha = 1, \beta = \int_0^1 |t| dt = \int_0^1 t dt = \frac{1}{2}$ and

when

$\alpha = -1, \beta = \int_0^{-1} |t| dt = -\int_{-1}^0 |t| dt = -\int_{-1}^0 t dt = -\frac{1}{2}$

$\therefore$ the points where the tangents are parallel to

the line $y = x$ are $\left(1, \frac{1}{2}\right), \left(-1, -\frac{1}{2}\right)$

The tangent at $\left(1, \frac{1}{2}\right)$ is $y - \frac{1}{2} = 1(x - 1)$,

i.e. $2x - 2y = 1$

The tangent at $\left(-1, -\frac{1}{2}\right)$ is $y + \frac{1}{2} = 1(x + 1)$,

i.e. $2x - 2y + 1 = 0$

44. Let $h(x) = (f(x))^{g(x)}$

$\Rightarrow \log(h(x)) = g(x)\{\log f(x)\}$

$\Rightarrow \frac{1}{h(x)} \cdot h'(x) = \frac{g(x)}{f(x)} \cdot f'(x) + \{\log f(x)\} g'(x)$

$\therefore h(x)$ is decreasing, if $\log$

$(f(x)) < 0 \Rightarrow f(x) < 1$

and $h(x)$ is increasing, if $\log$

$(f(x)) > 0 \Rightarrow f(x) > 1$

Hence : (b) and (d) are the correct answers.

APPLICATION OF DERIVATIVES

45. $f'(x) = \begin{cases} -1 & -2 < x < -1 \\ 1 & -1 \leq x < 0 \\ 0 & x = 0 \\ -\frac{1}{3}(1-x)^{-2/3} & 0 < x < 1 \\ \frac{1}{2\sqrt{x+1}} & x \geq 1 \end{cases}$

46. $\Rightarrow f'(x) = \frac{4x}{(x^2 + 1)^2}$

For max/min $f'(x) = 0 \Rightarrow x = 0$

$f''(x)|_{x=0} = +ve$

47. The maximum value of $f(x) = \cos x + \cos(\sqrt{2}x)$ is 2 which occurs at $x = 0$.

Also, there is no value of x for which this value will be attained again.

48. $f(2) = 18 \Rightarrow 8a + 4b + 2c + d = 18$ (i)

$f(1) = -1 \Rightarrow a + b + c + d = -1$ (ii)

$f(x)$ has local max. at $x = -1$

$\Rightarrow 3a - 2b + c = 0$ (iii)

$f'(x)$ has local min. at $x = 0 \Rightarrow b = 0$ (iv)

Solving (i), (ii), (iii) and (iv), we get result.

$\Rightarrow f(x) = \frac{1}{4}(19x^3 - 57x + 34)$

49. $\Rightarrow g'(x) = f(x) = \begin{cases} e^x, & 0 \leq x \leq 1 \\ 2 - e^{x-1}, & 1 < x \leq 2 \\ x - e, & 2 < x \leq 3 \end{cases}$

$\therefore g'(x) = 0 \Rightarrow e^{x-1} = 2$ or $x - e = 0$

$\Rightarrow x = 1 + 2 \ln 2$ or e

$\Rightarrow g''(x) = \begin{cases} e^x, & 0 \leq x \leq 1 \\ -e^{x-1}, & 1 < x \leq 2 \\ 1, & 2 < x \leq 3 \end{cases}$

50. For $f(x) = x \cos \frac{1}{x}, x \geq 1$

APPLICATION OF DERIVATIVES

$$f'(x) = x \cos\left(\frac{1}{x}\right) + \frac{1}{x} \sin\left(\frac{1}{x}\right) \rightarrow 1 \text{ for } x \rightarrow \infty$$

$$\text{also } f''(x) = \frac{1}{x^2} \sin\left(\frac{1}{x}\right) - \frac{1}{x^2} \sin\left(\frac{1}{x}\right) - \frac{1}{x^3} \cos\left(\frac{1}{x}\right)$$

$$\frac{1}{x^2} \sin\left(\frac{1}{x}\right) - \frac{1}{x^2} \sin\left(\frac{1}{x}\right) - \frac{1}{x^3} \cos\left(\frac{1}{x}\right) < 0 \text{ for } x \geq 1$$

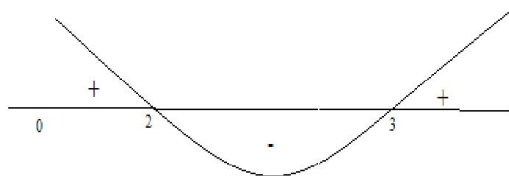
$\Rightarrow f'(x)$ is decreasing for $x \rightarrow \infty$

$\Rightarrow f'(x+2) < f'(x)$. Also

$$\lim_{x \rightarrow \infty} f(x+2) - f(x) = \lim_{x \rightarrow \infty} \left[(x+2) \cos \frac{1}{x+2} - x \cos \frac{1}{x} \right] = 2$$

$$\therefore f(x+2) - f(x) > 2 \forall x \geq 1.$$

51. $f'(x) = e^{x^2} (x-2)(x-3)$



ASSERTION & REASON QUESTIONS

52. counter example $f(x) = \frac{1}{x} \forall x \in (1, 2)$

$$\Rightarrow f'(x) = \frac{-1}{x^2} < 0, f''(x) = \frac{2}{x^3} > 0, \forall x \in (1, 2)$$

f is strictly increasing and $f'(x)$ is strictly decreasing in $(1, 2)$

INTEGER QUESTIONS

53. At $x = 0, y = 1$

Evaluate $\left. \frac{dy}{dx} \right|_{\text{at } x=0 \text{ \& } y=1}$

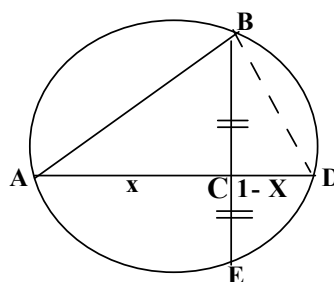
Find equation of tangent at $x = 0$ and $y = 1$.

54. Angle of intersection for $x > 1$ and $x < -1$ are similar because of similarity about y -axis.
for $x > 1$, point of intersection is $x = \sqrt{2}$.

Evaluate $\frac{dy}{dx}$ at $x = \sqrt{2}$ for $y = x^2 - 1$ and $y = 3 - x^2$.

55.

JEE ADVANCED - VOL - III



$$BC \times CE = AC \times CD \Rightarrow (BC)(CE) = x(1-x)$$

$$\therefore BC = \sqrt{x(1-x)} \Rightarrow \text{Area } \Delta = \frac{x\sqrt{x-x^2}}{2}$$

$$\Rightarrow \Delta^2 = \frac{x^3 - x^4}{2} \Rightarrow \frac{d\Delta^2}{dx} = \frac{3x^2 - 4x^3}{2}$$

If $\frac{d\Delta^2}{dx} = 0 \Rightarrow x = 3/4$ which is the point of maxima.

56. $f'(x) = 6(x-2)(x-3)$

so $f(x)$ is increasing in $(3, \infty)$

Also, $A = \{4 \leq x \leq 5\}$

$\therefore f_{\max} = f(5) = 7$

57. $f(0) = 1, f'(x) = 3x^2 + \frac{1}{2}e^{x/2} \Rightarrow f'(g(x))g'(x) = 1$

put $x = 0 \Rightarrow g'(1) = \frac{1}{f'(0)} = 2$

58. $g(x) = e^{f(x)}$ and $g^1(x) = e^{f(x)} \cdot f^1(x)$

$$g^1 = e^f [2010(x-2009)(x-2010)^2(x-2011)^3(x-2012)^4]$$

$$g^1(x) = 0 \Rightarrow x = 2009, 2010, 2011, 2012$$

at which $g^1(x)$ changes sign at

$$x = 2009, 2011$$

g has maximum at $x = 2009$ one point

59. $y - y_1 = m(x - x_1)$ is the equation of the tangent at (x_1, y_1)

y intercept $y_1 - mx_1 = x_1^3, \therefore x \frac{dy}{dx} - y = -x^3$

$$\frac{dy}{dx} - \frac{y}{x} = -x^2. \text{ the I.F. } = e^{\int -\frac{1}{x} dx} = \frac{1}{x}$$

The curve is $\frac{y}{x} = -x^2 \cdot \frac{1}{x} dx \Rightarrow \frac{y}{x} = -\frac{x^2}{2} + c$

$$f(1) = 1 \Rightarrow c = \frac{3}{2}. \text{ Hence}$$

$$f(x) = y = -\frac{x^3}{2} + \frac{3}{2}x, \quad f(-3) = 9$$

$$60. \quad f(x) = x^4 - 4x^3 + 12x^2 + x - 1$$

clearly $f(0) = -1 < 0 \Rightarrow$ at least two real roots

$f'(x) = 4x(x^2 - 3x + 7)$ dont have all real roots

$\therefore f(x)=0$ has only two real roots.

$$61. \quad p'(x) = 3k\{(x-1)(x-3)\} \\ = 3k\{x^2 - 4x + 3\}$$

$$p(x) = k\{x^3 - 6x^2 + 9x\} + c$$

$$p(1) = 6 \Rightarrow 4k + c = 6, \quad p(3) = 2 \Rightarrow c = 2$$

$$\Rightarrow k = 1 \quad \therefore p'(0) = 9k = 9$$

COMPERHENSION TYPE QUESTIONS

PASSAGE-1:- 62 - 64

$$62. \quad \frac{dy}{dx} = \frac{1-9t^2}{-6t} = \tan \theta \Rightarrow 9t^2 - 6 \tan \theta t - 1 = 0 \\ \Rightarrow 3t = \tan \theta \pm \sec \theta \\ \tan \theta + \sec \theta = 3t.$$

$$63. \quad P(-2, 2) \Rightarrow t = -1 \Rightarrow \left. \frac{dy}{dx} \right|_{t=-1} = -\frac{4}{3}$$

$$\text{Equation of the tangent } y - 2 = -\frac{4}{3}(x + 2)$$

$$\Rightarrow t - 3t^3 - 2 = -\frac{4}{3}(1 - 3t^2 + 2)$$

$$\Rightarrow 9t^3 + 12t^2 - 3t - 6 = 0$$

$$\Rightarrow 3t^3 + 4t^2 - t - 2 = 0$$

$$(3t^2 + t - 2)(t + 1) = 0$$

$$\Rightarrow (3t - 2)(t + 1) = 0 \Rightarrow t = \frac{2}{3} \Rightarrow Q = \left(-\frac{1}{3}, -\frac{2}{3}\right)$$

$$64. \quad \left. \frac{dy}{dx} \right|_{t=2/3} = \frac{3}{4}$$

$$m_{po} m_Q = -1 \Rightarrow \text{angle } 90^\circ$$

PASSAGE-2:- 65 - 66

$$g'(x) = f'(\sin x) \cos x - f'(\cos x) \sin x$$

$$\Rightarrow g''(x) = -f''(\sin x) \sin x + \cos^2 x \quad f''(\sin x)$$

(as it is given

$$f'(\sin x) = f'(\cos(\pi/2 - x)) < 0$$

$$f''(\sin x) = f''(\cos(\pi/2 - x)) > 0$$

$\Rightarrow g'(x)$ is increasing in $(0, \pi/2)$. Also

$$g'(\pi/4) = 0 \Rightarrow g'(x) > 0 \forall x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right) \text{ and}$$

$$g'(x) < 0 \forall x \in (0, \pi/4).$$

Thus $g(x)$ is decreasing in $(0, \pi/4)$.

PASSAGE-3:- 67 - 69

$$f(x)f'(x) \geq 0 \text{ in } [a, c] \text{ and } \leq 0 \text{ in } [c, b]$$

$$\therefore f(c) = -\sqrt{2} \text{ and } f(x)f'(x) \neq 0 \text{ in } (a, c) \cup (c, b)$$

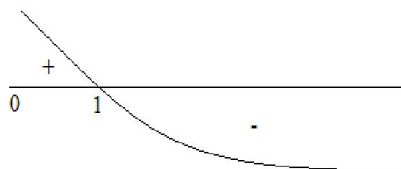
$$\Rightarrow f(a), f(b) \leq 0.$$

$f(0)$	$f(b)$	$f(c)$
0	0	$-\frac{3}{2}$
0	-1	$-\frac{3}{2}$
0	$-\sqrt{2}$	$-\frac{3}{2}$
-1	-1	$-\frac{3}{2}$
-1	$-\sqrt{2}$	$-\frac{3}{2}$
-1	$-\sqrt{2}$	$-\frac{3}{2}$

APPLICATION OF DERIVATIVES

PASSGE-4 :- 70 - 71

$$g'(x) = \left(\frac{2(x-1)}{x+1} - \ln x \right) \left[(1-x)^2 \sin^2 x + x^2 \right]$$



$g(x)$ decreases in the interval $(1, \infty)$

$$\left((1-x)^2 \sin^2 x + x^2 \right) + 2x = 2(1+x^2)$$

$$(1-x)^2 \sin^2 x = (1-x)^2 + 1$$

$1 + (1-x)^2 \cos^2 x = 0$, no real value of x exist.

PASSGE-5 :- 72-74

$$72. \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$(a \cos \theta, b \sin \theta) \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

$$p = \frac{1}{\sqrt{\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}}}$$

$$\Rightarrow \frac{1}{p^2} = \frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}$$

$$r^2 = a^2 \cos^2 \theta + b^2 \sin^2 \theta$$

$$\Rightarrow \frac{r^2}{a^2 b^2} = \frac{\cos^2 \theta}{b^2} + \frac{\sin^2 \theta}{a^2}$$

$$\therefore \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} - \frac{r^2}{a^2 b^2}$$

$$73. \quad x = a \cos^3 \theta, y = a \sin^3 \theta$$

$$(a \cos^3 \theta, a \sin^3 \theta)$$

JEE ADVANCED - VOL - III

$$\frac{dy}{dx} = \frac{a 3 \sin^2 \theta \cdot \cos \theta}{a 3 \cos^2 \theta (-\sin \theta)} = -\frac{\sin \theta}{\cos \theta}$$

$$y - a \sin^3 \theta = -\frac{\sin \theta}{\cos \theta} (x - a \cos^3 \theta)$$

$$\frac{x}{\cos \theta} + \frac{y}{\sin \theta} = a$$

$$p = \frac{a}{\sqrt{\frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}}} = a \cos \theta \sin \theta$$

$$r^2 = a^2 \cos^6 \theta + a^2 \sin^6 \theta =$$

$$a^2 \left((\cos^2 \theta)^3 + (\sin^2 \theta)^3 \right)$$

$$= a^2 (\cos^4 \theta - \cos^2 \theta \sin^2 \theta + \sin^4 \theta)$$

$$= a^2 (1 - 3 \cos^2 \theta \sin^2 \theta)$$

$$r^2 = a^2 - 3p^2$$

$$74. \quad c^2 (x^2 + y^2) = x^2 y^2$$

$$\frac{1}{x^2} + \frac{1}{y^2} = \frac{1}{c^2} \quad (c \sec \theta, c \operatorname{cosec} \theta)$$

$$-\frac{2}{x^3} - \frac{2}{y^3} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\left(\frac{y^3}{x^3} \right) = -\left(\frac{\operatorname{cosec}^3 \theta}{\sec^3 \theta} \right)$$

$$y - \operatorname{cosec} \theta = -\frac{\operatorname{cosec}^3 \theta}{\sec^3 \theta} (x - \sec \theta)$$

$$y \sin^3 \theta - c \sin^2 \theta = -x \cos^3 \theta + c \cos^2 \theta$$

$$x \cos^3 \theta + y \sin^3 \theta = c$$

$$r = c^2 \sec^2 \theta + c^2 \operatorname{cosec}^2 \theta$$

$$= \frac{c^2}{\sin^2 \theta \cos^2 \theta} \Rightarrow \frac{1}{r^2} = \frac{\sin^2 \theta \cos^2 \theta}{c^2}$$

$$p = \frac{c}{\sqrt{\cos^6 \theta + \sin^6 \theta}}$$

$$\frac{1}{p^2} = \frac{\cos^6 \theta + \sin^6 \theta}{c^2}$$

$$\frac{1}{p^2} = \frac{\cos^4 \theta - \cos^2 \theta \sin^2 \theta + \sin^4 \theta}{c^2}$$

$$\frac{1}{p^2} = \frac{1 - 3 \cos^2 \theta \sin^2 \theta}{c^2}$$

$$\Rightarrow \frac{1}{p^2} + \frac{3}{r^2} = \frac{1}{c^2}$$

PASSGE :- 75-77

Let $P_1(t_1, t_1^3)$, $P_2(t_2, t_2^3)$, $P_3(t_3, t_3^3)$

Be points on the curve

equation of tangent at P_1

$$y - t_1^3 = 3t_1^2(x - t_1)$$

Given it meets curve at P_2

$$\Rightarrow t_2^3 - t_1^3 = 3t_1^2(t_2 - t_1)$$

$$\Rightarrow t_1^2 + t_2^2 + t_1 t_2 = 3t_1^2 \Rightarrow t_2^2 - 2t_1^2 + t_1 t_2 = 0$$

$$\Rightarrow t = -2t_1 \quad (\because t_1 \neq t_2)$$

$$t_3 = -2t_2, \quad t_4 = -2t_3 \text{ and so on}$$

$\therefore$ Abcissa are in G.P with $Cr - 2$

ordinates are in G.P with $Cr - 8$

MATRIX- MATCH QUESTIONS

$$78. \quad g(x) = \begin{cases} f(x) & -2 \leq x < -1 \\ f(-1) & -1 \leq x < 0 \\ f(0) & 0 \leq x < 1 \\ f(x) & 1 \leq x \leq 3 \end{cases}$$

APPLICATION OF DERIVATIVES

$$f(x) = \begin{cases} x^2 + 2x & -2 \leq x < -1 \\ -1 & -1 \leq x < 0 \\ 0 & 0 \leq x < 1 \\ x^2 - 2x & 1 \leq x \leq 3 \end{cases}$$

(a) $f(x)$ not continuous at $x = 0$

(b) $g(x)$ not continuous at $x = 0, 1$ and not differentiable at $0, 1$.

(c) No point exist for local extrema

(d) Absolute maxima occurs at $x = 3$

SUBJECTIVE QUESTIONS

79. Let, $f(x) = 2\sin x + 2\tan x - 3x$

$$\Rightarrow f'(x) = 2\cos x + 2\sec^2 x - 3$$

since, $0 \leq x < \pi/2$, $f'(x) > 0$

Thus, $f(x)$ is increasing, when $x \geq 0$,

$$f(x) \geq f(0)$$

80. Given $f(x)$

$$\left\{ \begin{array}{ll} -x^3 + \frac{(b^3 - b^2 + b - 1)}{(b^2 + 3b + 2)} & , \quad 0 \leq x \leq 1 \\ 2x - 3 & , \quad 1 \leq x \leq 3 \end{array} \right\} \text{ is}$$

smallest at $x = 1$

$$f'(x) = \begin{cases} -3x^2 & 0 < x < 1 \\ 2 & 1 < x < 3 \end{cases}$$

So, $f(x)$ is decreasing on $[0, 1)$ and increasing on $(1, 3]$

Here, $f(1) = -1$ is the smallest value at $x = 1$.

$\therefore$ its smallest value occur as,

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (-x^3) + \frac{(b^3 - b^2 + b - 1)}{b^2 + 3b + 2}$$

in order this value is not less than -1 , we must have

$$\frac{b^3 - b^2 + b - 1}{b^2 + 3b + 2} \geq 0 \Rightarrow \frac{(b^2 + 1)(b - 1)}{(b + 1)(b + 2)} \geq 0$$

$$\therefore b \in (-2, -1) \cup [1, \infty)$$

81. It is given that $y = ax^3 + bx^2 + cx + 5$ touches x-axis at $P(-2, 0)$ which implies that x-axis is tangent at $(-2, 0)$ and the curve is also passes through $(-2, 0)$. The curve cuts y-axis at $(0, 5)$ and gradient at this point is given 3 therefore at $(0, 5)$ slope of the tangent is 3.

APPLICATION OF DERIVATIVES

Now, $\frac{dy}{dx} = 3ax^2 + 2bx + c$

Since x-axis is tangent at $(-2, 0)$ therefore

$$\left. \frac{dy}{dx} \right|_{x=-2} = 0 \Rightarrow 0 = 3a(-2)^2 + 2b(-2) + c$$

$$\Rightarrow 0 = 12a - 4b + c \dots (i)$$

again slope of tangent at $(0, 5)$ is 3

$$\Rightarrow \left. \frac{dy}{dx} \right|_{(0,5)} = 3 \Rightarrow 3 = 3a(0)^2 + 2b(0) + c$$

$$\Rightarrow 3 = c \dots (ii)$$

Since, the curve passes through $(-2, 0)$, we get

$$0 = a(-2)^3 + b(-2)^2 + c(-2) + 5$$

$$0 = -8a + 4b - 2c + 5 \dots (iii)$$

Hence, $a = -1/2$, $b = -3/4$ and $c = 3$.

82. $f(x)$ is continuous at $x = 0$.

$$\text{Also, } f'(x) = \begin{cases} 1 \cdot e^{ax} + axe^{ax} & , \text{ if } x < 0 \\ 1 + 2ax - 3x^2 & , \text{ if } x > 0 \end{cases}$$

$$Lf'(0) = Rf'(0) = 1 \Rightarrow f'(0) = 1$$

$$\text{Hence, } f'(x) = \begin{cases} (ax+1)e^{ax} & , \text{ if } x < 0 \\ 1 & , \text{ if } x = 0 \\ 1 + 2ax - 3x^2 & , \text{ if } x > 0 \end{cases}$$

$$f''(x) = \begin{cases} ae^{ax} + a(ax+1)e^{ax} & , \text{ if } x < 0 \\ 2a - 6x & , \text{ if } x > 0 \end{cases}$$

$$\text{and } Lf''(0) = 2a \text{ and } Rf''(0) = 2a$$

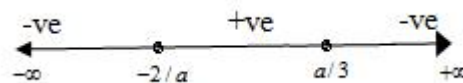
$$\text{Therefore, } Lf''(0) = Rf''(0) = 2a$$

$$\text{Hence, } f''(x) = \begin{cases} a(ax+2)e^{ax} & , \text{ if } x < 0 \\ 2a & , \text{ if } x = 0 \\ 2a - 6x & , \text{ if } x > 0 \end{cases}$$

$$\text{for the critical points } f''(x) = 0 \Rightarrow x = \frac{-2}{a} \text{ if}$$

$$x < 0 \text{ and } x = \frac{a}{3} \text{ if } x > 0$$

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Hence, $f(x)$ increases on $\left[\frac{-2}{a}, \frac{a}{3} \right]$.

83. Let $f(x) = \sin(\tan x) - x$

$$f'(x) = \cos(\tan x) \cdot \sec^2 x - 1$$

$$= \cos(\tan x) \{1 + \tan^2 x\} - 1$$

$$= \tan^2(\cos(\tan x)) + \cos(\tan x) - 1$$

$$> \tan^2 x \cos(\tan x) - \frac{\tan^2 x}{2}$$

$$\left\{ \because 0 \leq \tan x \leq 1 < \frac{\pi}{3} \right\}$$

$$f'(x) > \tan^2 x \left[\cos(\tan x) - \frac{1}{2} \right]$$

$$> \tan^2 \{ \cos(\tan x) - \cos(\pi/3) \} > 0$$

$$\Rightarrow f(x) \text{ is increasing function } \forall x \in [0, \pi/4]$$

$$\text{as } f(0) = 0 \Rightarrow f(x) \geq 0 \quad \forall x \in [0, \pi/4]$$

$$= \sin(\tan x) \geq x$$